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Calcination of $CaCO_3$ and the subsequent sintering/densification of CaO directly visualized with in-situ SEM

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HIGHLIGHTS

- Explored in-situ SEM for dynamic heating on $CaCO_3$ during calcination.
- MEMS heating was utilized for uniformity and reproducibility.
- Observed microstructural changes including meso/macropores and cracks.
- Onset of grain formation and merging observed during sintering.
- Findings enhance calcination process optimization for efficiency.

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ABSTRACT

This work investigates the possibility and potential insights gained from in-situ scanning electron microscopy (SEM) experiments focused on the dynamic heating of $CaCO_3$ for calcination processes. MEMS chips were used to facilitate uniform heating, precise control of heating parameters and reproducibility, providing a controlled environment for comprehensive experimentation. This work focuses on the microstructure transformation, with particular emphasis on the meso- and macropores, grain formation, and the sintering behavior of these particles. Influence of heating rates and complex interplay of temperature and microstructural transformations was revealed. The results obtained may be useful for optimization and microstructural-based design of the calcination procedures for complex process engineering in various applications.

1. Introduction

Thermal decomposition (calcination) of a natural limestone ($CaCO_3$) is a frequently employed process where the resulting lime (CaO) is used as an ingredient in the cement [1], paper [2], paints/coatings [3,4] and plastics [5,6]. Calcination is the process of heating a substance to a high temperature in the production of various inorganic materials, including calcium carbonate ($CaCO_3$, resulting in a chemical reaction that changes its physical and/or chemical properties. In thermal calcination process given in Eq (1), $CaCO_3$ decomposes to form calcium oxide (CaO) and carbon dioxide (CO_2) between 500 °C and 900 °C [7]. The cement industry contributes to approximately 7–8% [8,9] of the global CO_2 emission while the specific emission attributed to the decomposition of limestone used in this industry alone is estimated to be around 4% [9].

The CaO formed during the thermal decomposition of $CaCO_3$ contains some minor impurities, is highly reactive and can react with other materials, leading to the formation of new phases. Causticizing process in the kraft pulp mills uses CaO to convert Na_2CO_3, resulting in the formation of $CaCO_3$. The $CaCO_3$ is subsequently separated, washed and dried to reuse as lime mud in the kiln. The $CaCO_3$ is also extensively used as a sorbent precursor to produce CaO in the calcium looping process for carbon capture in the fossil fuel-based plants and for thermochemical energy storage devices [10–12].

$$CaCO_3 \Leftrightarrow CaO + CO_2 \quad \Delta H = +177.8\ kJ\ mol^{-1} \qquad \text{Equation (1)}$$

Calcination of limestone is routinely carried out in the rotary lime kiln and is a complicated process of reaction between chemical and physical changes appearing in the kiln at high temperature. During the

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calcination process, morphological changes e.g. pores start appearing on the surface of the material and has been attributed due to a difference in the specific density (2.8 g/cm^3) of $CaCO_3$ and CaO (3.3 g/cm^3) [7,13]. The microstructural changes that appeared during the calcination process have been studied ex-situ using conventional experimental techniques such as mercury porosimetry, gas adsorption and thermogravimetric analysis [10,14–16]. These ex-situ studies have several experimental limitations. In-situ liquid cell transmission electron microscopy experiments have been reported in the literature to understand the time evolution of $CaCO_3$ synthesis from nanodroplet precursor to crystalline calcite [17] and real time observation of mineralization processes of $CaCO_3$ [18]. In recent years, several in-situ studies have also been conducted to further understand the thermal behavior of $CaCO_3$ using X-ray diffraction (XRD), (small/wide angle X-ray scattering (SAXS/WAXS) techniques. Matteo et al. [7] has conducted a study on the thermal behavior of porous materials using an in-situ SAXS/WAXS model to calculate the micro texture properties such as porosity, surface area, etc. of calcined $CaCO_3$ powder. Santiago et al. [10] has carried out a comparative study to analyses the calcination and carbonation cycles of natural limestone and dolomite by means of an in-situ X-ray diffraction. In another study, Valverde et al. [19] conducted calcination experiments under several different environments (CO_2, N_2, O_2, He and mixture). They identified the potential high operating cost associated with maintaining a temperature of 930 °C to achieve a full calcination at a residence time of (typically <10 min) and to generate highly concentrated CO_2 under high partial pressure of CO_2. However, based on the findings of the in-situ X-ray study, the authors proposed that the utilization of a small concentration of He could substantially boost the calcination rate of limestone under high partial pressure of CO_2. This, in turn, could decrease the residence time (<10 min) required at temperatures slightly above 900 °C.

In-situ Scanning Electron Microscope (SEM) thermal heating is a technique that allows the observation of morphological changes in materials in real-time while they are subjected to high temperatures. In this technique, a sample is placed inside the SEM chamber and is heated to a desired temperature while being observed through the SEM imaging system. The advantage of in-situ SEM thermal heating is that it provides a direct observation of the morphological changes in materials at high temperatures, making it possible to understand the underlying mechanisms of these changes.

Sintering is also the process of heating the material at high temperatures in the presence or absence of a sintering agent to cause densification and/or formation of new compounds. CO_2 diffusing out from the $CaCO_3$ particles during the calcination process, accelerates the sintering phenomenon and continues even after the calcination [20]. The high temperature range for sintering of CaO typically starts from 600 °C to 1000 °C and can go up to 1150 °C [21]. Sintering mechanisms of CaO particles have been studied and reported in the presence of water vapors [22] and CO_2/O_2 [23–25]. The studies have reported the impact of atmospheric conditions on pore size, pore volume and the carbonation conversion rate. However, there is still no consensus on the underlying mechanism involving sintering.

The objective of this study was to investigate the in-situ experimentation possibilities of calcination, aiming to provide valuable insights into this critically important phenomenon. In particular, the focus was on observing the dynamic morphological changes that appear during thermal calcination and sintering experiments. To achieve this, we employed in-situ SEM with micro-electromechanical system (MEMS) based heating, aiming to capture real-time images. Additionally, the heating experiments were designed to comprehensively analyze the influence of varying heating rates. This approach allows to gain an understanding of the complex process associated with calcination and sintering phenomena.

2. Materials and methods

2.1. Materials

$CaCO_3$ powder particles with an average particle size of 14 μm, density of 2.8 g/cm^3 and a purity of 98.5% was obtained from Emsure®. Based on powder X-ray diffraction (XRD) analysis, the sample is crystalline and no crystalline phases other than calcite ($CaCO_3$) could be identified (see supplementary material, Fig. S1a). The powder was grinded using the mortar and pestle before dispersing onto the ThermoFisher Scientific NanoEX i/v MEMS chip [26]. Individually calibrated MEMS chips have a uniform temperature distribution over the circular area of 100 μm, high thermal stability, accuracy and fast heating can attain a temperature of 1200 °C with a ramp rate of 10000 °C/s.

2.2. In-situ SEM heating experiments

The MEMS chip based in-situ heating experiments were performed in the Apreo 2S Field Emission SEM equipped with a Thermo Scientific UltraDry Energy Dispersive X-ray spectroscopy (EDS) detector. MEMS chip with a dispersed $CaCO_3$ powder particles was mounted in the μheater holder (see Fig. 1d.). The μheater holder was then mounted onto the ThermoFisher modified multipurpose holder that supports heater contacting and also allows stage rotation. Fig. 1 shows the structure of the MEMS chips, circular heating area, electrical biasing and the deposited $CaCO_3$ powder. The joule effect-based heating is generated when a current is passed through the membranes connected to the circular heating area of the MEMS chip. A series of heating experiments were performed under high vacuum in-steps and the temperature ramp rate from 40 °C to 1000 °C was the only variable parameter. The heating experiments are classified based on the selection of temperature ramp rates from 40 °C to 1000 °C in this paper: ramp rate of 0.1 s (flash-heating), 100 s (rapid-heating) and 1000 s (fast-heating). The residence time of 20–25 min was maintained at 1000 °C. Since calcination of $CaCO_3$ in a CO_2 free atmosphere occurs at lower temperature then 1000 °C the resulting particle composes hereafter of CaO which has been verified with XRD (Fig. S1b). The temperature was then raised from 1000 °C to 1200 °C and maintained for a residence time of 90–122 min.

After heating experiments, the sample was cooled down to 40 °C. Pre- and post-heating SEM-EDS point analysis was performed at 40 °C to determine the chemical composition. Secondary electron SEM imaging was performed with a conventional Everhart-Thornley Detector (ETD) to follow the morphological changes before, after, and during the heating experiment. The SEM chamber maintained a pressure of around 10^{-4} Pa. The imaging was performed at 2 keV and the EDS was performed at 20 keV.

3. Results and discussion

The aim of these experiments is to observe and monitor the dynamic changes that appeared during the in-situ SEM thermal heating and to identify the intermediate and final products that result from the calcination and sintering processes.

3.1. Microstructure of $CaCO_3$

Fig. 2a presents the SEM micrograph of a typical virgin $CaCO_3$ powder particles selected for the flash-heating experiment. No porosity or cracks were observed on the smooth surface of the selected particles before heating. Chemical composition of the $CaCO_3$ powder particles before heating was determined using the SEM-EDS.

3.2. Flash-heating experiment

The flash-heating was achieved by utilizing the MEMS chip and the temperature was raised from 40 °C to 1000 °C in 0.1 s. Immediately after

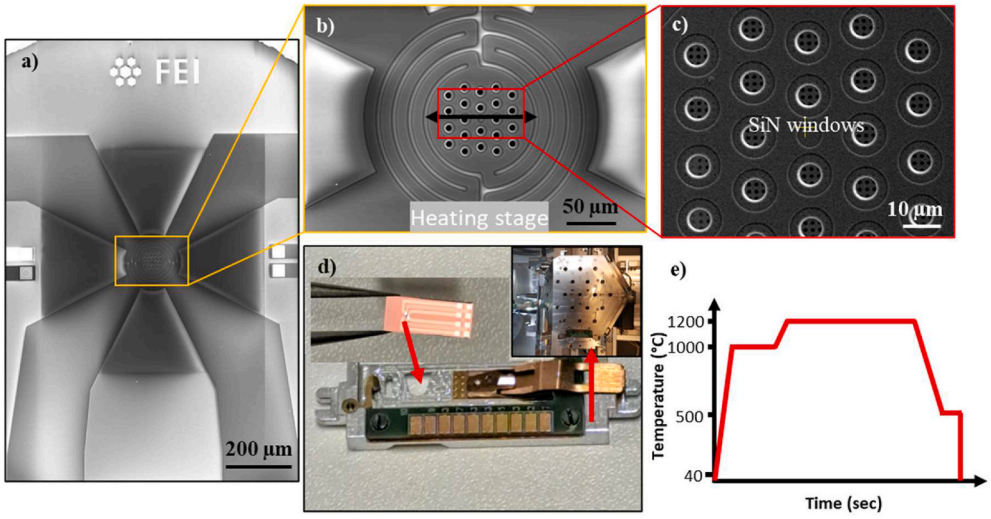


Fig. 1. (a) overview of the MEMS chip (b) The heating area of the NanoEX i/v MEMS (c) silicon nitride (SiN) based heating windows (d) photograph of the MEMS chip and the MEMS holder, inset on the top right showing the Multipurpose heating holder, (e) schematic of the temperature ramp during the heating experiments. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

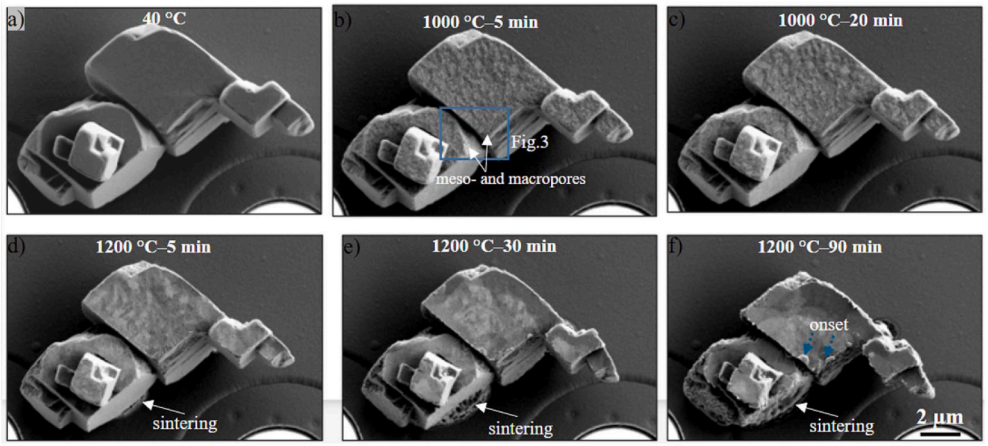


Fig. 2. Secondary electron SEM micrographs of the $CaCO_3$ powder particles subjected to flash heating, where the temperature is ramped from 40 °C to 1000 °C in 0.1 s virgin state of the particles (a); (b) and (c) show the particles surface morphology after 5 and 20 min at 1000 °C. The early stage of microstructural evolution at 1200 °C is shown in (d–f) corresponds to residence time of 5 min (d), 30 min (e) and 90 min (f). Scale marker in (f) is applicable to all the presented images.

the flash-heating at 1000 °C, morphological changes such as emergence and growth of meso- and macropores on the particles surface, occur during the residence time at 1000 °C (see Figs. 2 and 3). These morphological changes appeared during calcination and sintering have important implications for the properties and uses of resulting CaO, including its strength and reactivity [24,27–30]. Fig. 2 illustrates the different stages during the in-situ heating experiment. Meso- and macropores begin to form on the particle's surface as they reach a temperature of 1000 °C. The high magnification image in Fig. 3 illustrates the meso- and macropores that have developed on the surface of the powder particles after 20 min at 1000 °C. Morphological changes and microstructural evolution were continued at 1200 °C. The initiation of grain formation was promptly observed upon reaching a temperature of 1200 °C. The formation and growth of new grains in this process have been attributed to the sintering phenomenon. The grain growth persists throughout the residence time. In Fig. 4a, the apparent microstructural features on the surface of the particles were observed characterized by the appearance of meso- and macropores as highlighted by the red circles. In Fig. 4b, prolonged residence time of 120 min at 1200 °C leads to pronounced coalescence of surface domain leading to the onset of grains growth during sintering. The sintering process starts immediately after the temperature of 1200 °C is reached as indicated in Fig. 2d–f.

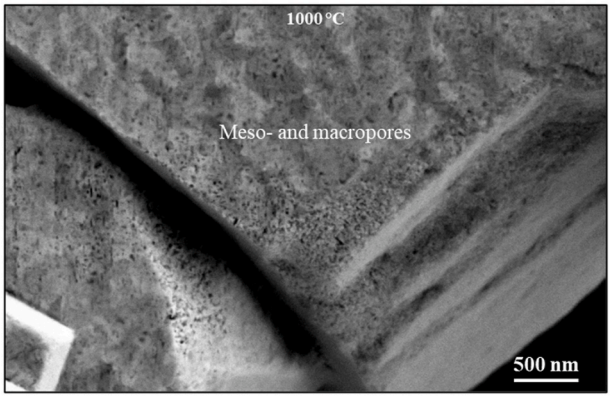


Fig. 3. Secondary electron high magnification image of Fig. 2(b) showcasing the meso- and macropores formed on the surface of the powder particles after being exposed at 1000 °C for 16 min.

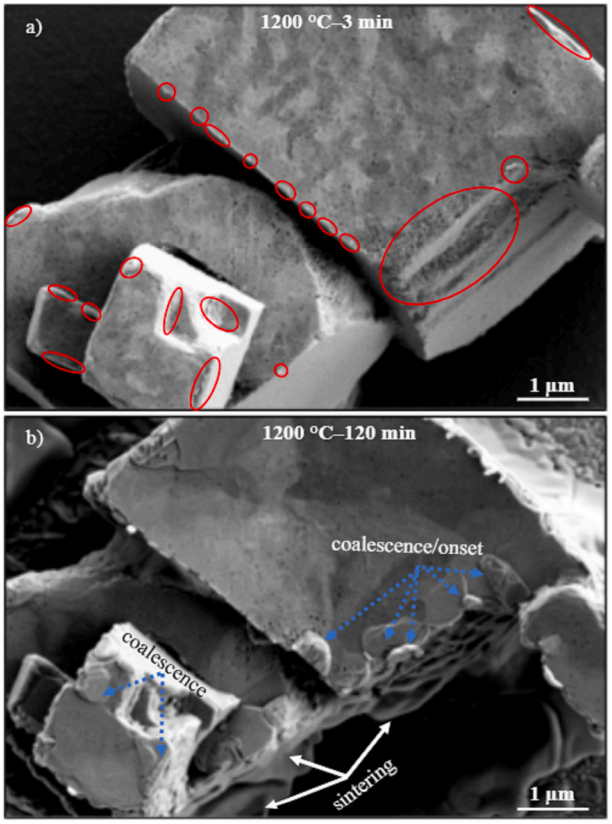


Fig. 4. Secondary electron SEM micrographs show the grain growth that occurred at 1200 °C. Early-stage microstructural changes characterized by the appearance of meso- and macropores and are highlighted by red circles in (a), (b) after residence time of 120 min, pronounced coalescence leads to the grain growth (marked with dotted blue arrows) during sintering, sintered regions are highlighted by white arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.3. Rapid-heating experiment

Rapid heating was achieved by increasing the temperature from 40 °C to 1000 °C in 100 s. In this experiment, a cluster of small $CaCO_3$ particles (Fig. 5a) situated on the heating area of the MEMS chips was chosen. Morphological transformations start appearing on the surface of the particles and cracks were observed on the surface during the residence time at 1000 °C. The illustration of cracks propagation on the surface of the particles is presented in Fig. 5b, c. However, no cracks were observed during the flash-heating experiments. The sintering process was observed immediately with the formation of small grains initiating as soon as the temperature was raised from 1000 °C to 1200 °C. At such a high a temperature, meso- and macropores were observed in addition to sintering. The formation of new grains, their growth and sintering during the residence time at 1200 °C is illustrated in Fig. 5d–f.

3.4. Fast-heating experiment

Fast-heating was achieved by raising the temperature from 40 °C to 1000 °C in 1000 s. $CaCO_3$ particles located in two distinct regions on the MEMS chip were selected for the experiment. Two distinct regions on the MEMS heater (particles in Figs. 6a and 7a) were selected and monitored during the experiment. Similar to the previous heating experiments, the morphological changes began to appear on the surface of the particles at 1000 °C. This includes the cracks propagation and the formation of meso- and macropores on the surface of the particles, as shown in Figs. 6b and 7b. The onset of grain formation was observed as soon as the temperature raised to 1200 °C and is presented in Fig. 6c and 7c and the grain growth continues during the residence time at 1200 °C. Nevertheless, in the region depicted in Fig. 7b, the sintering was evident at 1000 °C along with the initiation of grain growth and persisted as the temperature increased to 1200 °C. The formation of large grains and sintering was observed in both selected regions during the residence time at 1200 °C as illustrated in Fig. 6d and 7d.

3.5. Slow-heating experiment

The slow heating experiments were carried out with a ramp rate of 2.5 °C/s and the temperature was increased from 40 °C to 500 °C. The morphological changes and the initiation of meso- and macropores on the particles surface were observed after a residence time of 2 min at 500 °C, as shown in Fig. 8b. The formation of pores at 500 °C confirms the commencement of calcination. The presence and development of meso- and macropores on the particles surface were observed between 500 °C and 1000 °C Fig. 8 ((b–f)). The sintering was observed at 1200 °C, leading to the development of surface grains with discernible boundaries during sintering which are highlighted in Fig. 8h.

4. Discussion

The morphological changes that occur during the thermal decomposition of $CaCO_3$ were observed in real time using in-situ SEM thermal heating. $CaCO_3$ undergoes a series of microstructural transformations, including calcination (formation of CaO), pores formation, grain growth and sintering. In this in-situ SEM work, the onset of grain growth refers to the initial appearance of surface domains with discernible boundaries and coalescence that are distinguishable from the virgin particle surface at 40 °C, while grain growth describes the subsequent increase in the size and stability of these surface features during continued heating. Sintering is identified by progressive reduction of pores and coalescence of features at high temperature. These terms are used here to describe the morphological based observations. The observed morphological changes can be influenced by several factors, including the temperature, heating rate and residence time. These transformations primarily appear as changes in surface morphology and apparent grain size, which can significantly impact the properties and performance of the material. The reproducibility of the results was confirmed by repeating the heating experiments presented in section 3.

4.1. Formation of meso- and macropores and cracks

Meso- and macropores along with cracks were observed to develop during the heating experiments conducted, irrespective of heating conditions applied. The mechanism of meso- and macropores and cracks formation during thermal calcination process has been attributed to the volume change [7,31,32] that occurs during the decomposition of $CaCO_3$ to CaO and CO_2. During the thermal calcination process at high temperature, the CO_2 escapes from the $CaCO_3$ crystal structure, leading to formation of meso- and macropores. No significant difference in size of the particles was observed after heating but the meso- and macropores start appearing on the surface of the particles. The pore formation may be attributed to the thermal stresses at high temperature and difference in the specific density of $CaCO_3$ (2.8 g/cm^3) and CaO (3.3 g/cm^3) [7,23]. Several studies have reported the occurrence of thermally induced microcracks in limestone at different temperature thresholds, notably at 300 °C [33], 500 °C [34] and at 700 °C [35]. Kranz et al. [36] has identified different types of cracks such as intergranular (along the grain boundaries) and intergranular (originated from grain boundaries and crossing into one or more grains). Chen et al. [37] has reported that the appearance of microcracks, swelling and breaking of limestone can be attributed to the structural changes induced by heat expansion/thermal stress as a function of temperature. According to the findings of Eyad et al. [32] microcracks are attributed to the decrease in activation

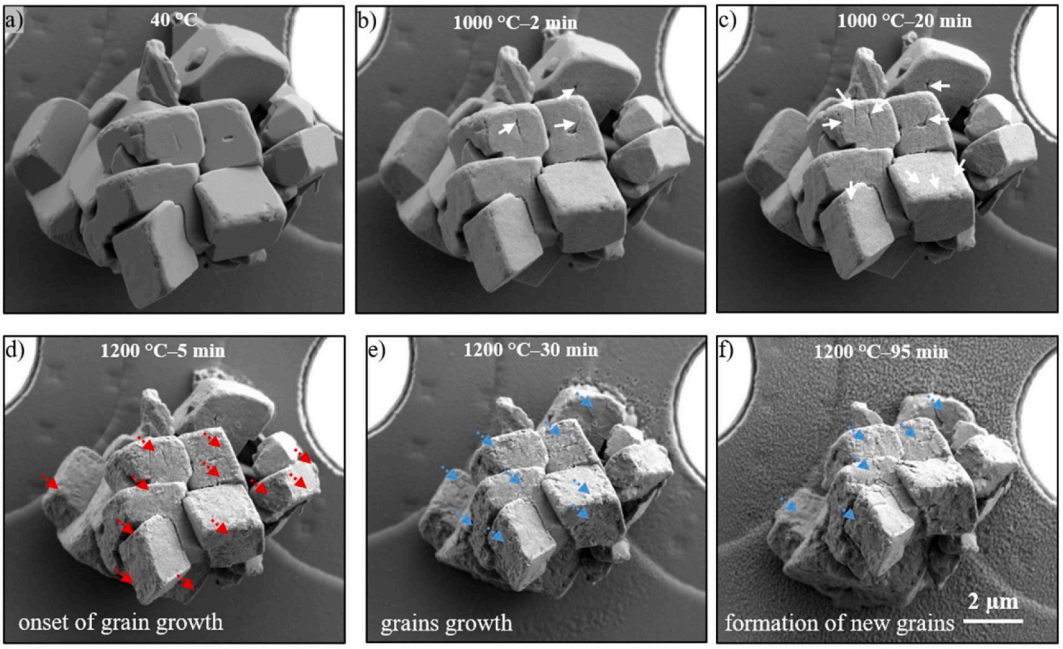


Fig. 5. Secondary electron SEM micrographs of $CaCO_3$ powder particles subjected to rapid-heating, where the temperature is ramped from 40 °C to 1000 °C in 100 s. (a) virgin state of the particles at 40 °C; (b) and (c) display the images at 1000 °C after 2 min and 20 min, respectively. The white arrows in (b) and (c) indicate thermally induced cracks formed on the particle surfaces. (d) The onset of grain growth (highlighted by dotted red arrows) and sintering observed after 5 min at 1200 °C. Continued residence at 1200 °C results in pronounced coalescence and grain growth after 30 min (e) and 95-min (f), indicative of formation of new grains during sintering. The scale marker in (f) is applicable to all the presented images. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

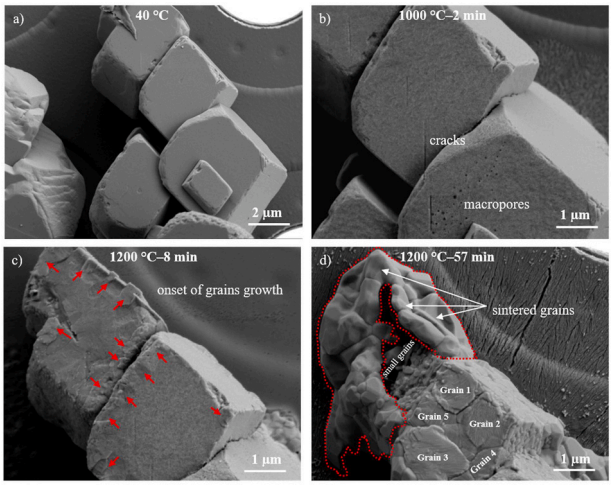


Fig. 6. SEM micrographs of $CaCO_3$ powder particles subjected to fast-heating, where the temperature was ramped from 40 °C to 1000 °C in 1000 s: (a) virgin state at 40 °C and (b) Particles surface morphology indicating pores and cracks formation after heating from 40 °C to 1000 °C in 1000 s. Onset of grain growth and sintering observed after residence time of 8 min at 1200 °C characterized by coalescence and apparent grain growth. (d) Prolonged residence at 1200 °C (after 57 min) results in pronounced grain growth with formation of large and more stable grains indicative of advanced sintering.

energy which in turn increases the surface-to-volume ratio and expedites the limestone calcination process especially when subjected to higher heating rates in an air environment. They also proposed that thermal stresses and pressure gradients between the interior and exterior of the particles may have the potential to initiate the microcracks observed on the surface [32]. Moreover, the presence of impurities or inherent defects in the $CaCO_3$ can also contribute to the formation of nanopores and microcracks during the thermal calcination [25,38]. The observed phenomenon of cracks and meso- and macropores formation in the heating experiments presented in section 3 is in accordance with earlier reported studies [16,39]. Notably, this is now observed for the first time through in-situ experiments.

4.2. Sintering, densification and formation of grains

Thermal calcination process induces the breakdown of the crystal structure of the $CaCO_3$, causing the densification of material at high temperatures. This process is known as sintering, the small particles of CaO cluster together resulting in the formation of grains of CaO. Throughout the sintering process, the system interfacial energy is reduced via diffusion, ultimately achieving a high density [40,41]. Ning et al. [40] proposed that densification and grain growth are the two main competing events which can alter the interfacial energy and interfacial area, respectively. They proposed that during the sintering of micron size ZnO ridges, the densification is through grain boundary diffusion, and the grain growth is through evaporation-condensation [40]. In all the heating experiments, during the sintering process at around 1200 °C, the meso- and macropores formed on the surface of the particles were observed to decrease and result in a more dense and homogenous material, as shown in Figs. 4–6. The observed findings in these heating experiments were consistent with previous studies [22,25,42]. The formation of new grains and grain boundaries, as observed during the thermal sintering process [Figs. 4–8], plays a crucial role in decreasing the overall system of energy. This reduction in system energy facilitates mass transport pathways and contributes to the pore elimination [40,43]. The SEM micrographs in.

Fig. 6 illustrate the aggregation of small grains leading to the formation of large grains in the calcined material. The sintered material exhibits improved mechanical properties [44,45]. The choice of sintering temperature and conditions depends on the desired final properties of the sintered product. In most of the conducted heating experiments, during the extended residence time at 1200 °C, the grains of CaO may

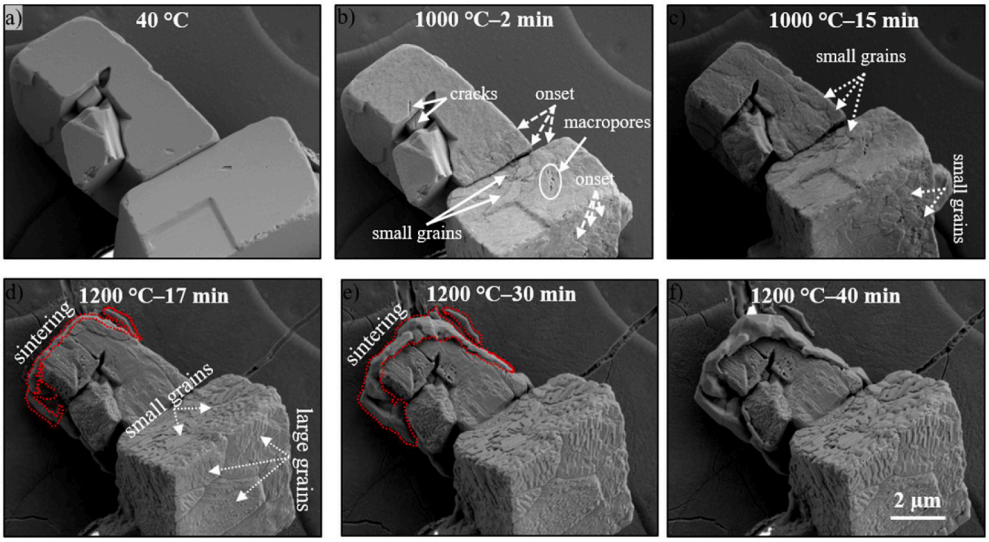


Fig. 7. Secondary electron SEM micrographs showing the $CaCO_3$ powder particles subjected to fast-heating, where the temperature was ramped from 40 °C to 1000 °C in 1000 s. (a) Virin state of the particles at 40 °C.(b and c) Particle surface morphology after heating from 40 °C to 1000 °C showing the formation of macropores, cracks and apparent onset of grain growth characterized by the coalescence of surface features. (d–f) Progressive grain growth and sintering resulting in the formation of large surface grains.

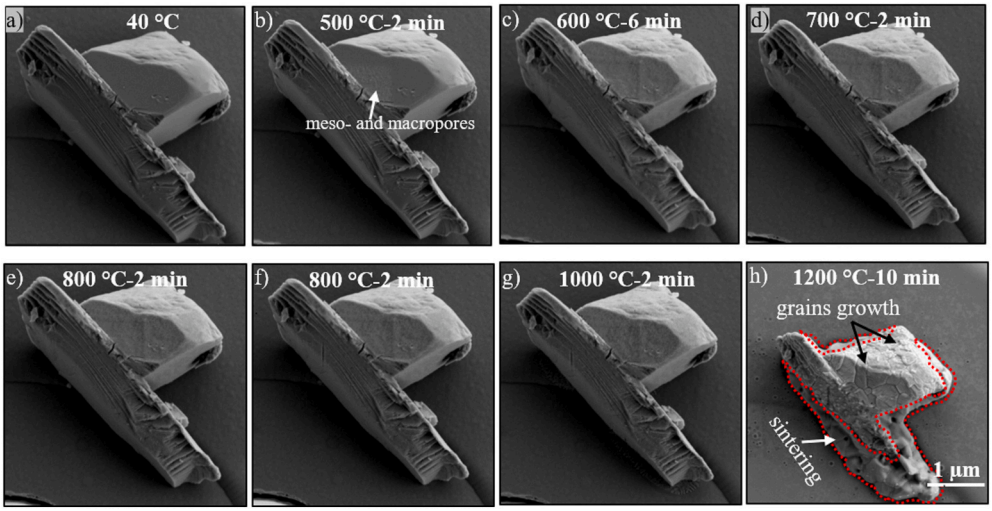


Fig. 8. Secondary electron SEM micrographs illustrating the different stages of surface morphological evolution of a $CaCO_3$ powder particle during the slow heating experiment: (a) Virgin state of the particle before heating at 40 °C. (b–f) initiation and development of meso- and macropores observed between 500 °C and 1000 °C. (h) formation of grains was observed along with the sintering during the residence time of 10 min at 1200 °C.

undergo morphological evolution and sintering-driven grain growth as evidenced by the formation of large grains with discernible boundaries in Fig. 6(c and d). The experimental findings of the present work are based on in-situ observations and provide strong evidence of the systematic temperature and time dependent morphological evolutions from pores to onset of grain formation and subsequently large grains associated with sintering. In-situ SEM has certain limitations when it comes to quantitative confirmation of grain growth, orientation and recrystallization mechanism. This may require certain complementary techniques such as high resolution XRD and electron back-scattered diffraction, etc.

5. Conclusion and future work

- This work successfully explores the potential of in-situ SEM for dynamic heating experiments on $CaCO_3$ during calcination process.
- MEMS based heating has been utilized for uniform heating and ensuring experiment reproducibility.
- Microstructural transformations, particularly meso- and macropores and thermal cracks development, were observed during the calcination process.
- On-set of grain formation, merging and formation of large grains was observed during the sintering of CaO particles during the residence time at 1000 °C and 1200 °C.
- The results obtained may improve the understanding and practical implications for the optimization and customization of calcination process for enhanced efficiency in various industrial applications.
- Future in-situ heating experiments will be designed to investigate the influence of various environments including air, O_2 and CO_2 on real limestone samples within a microreactor. Additionally, high temperature EDS analysis, high resolution EBSD and XRD will be performed to differentiate the chemical composition of the particles during the complex processes of calcination and sintering.

CRediT authorship contribution statement

Imran Hanif: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Henrik Wiinikka:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matchemphys.2026.132296.

Data availability

Data will be made available on request.

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